Request for Quotation 190184
Insourcing Solenoid Armature

19.09.2024
A07591000B / Bosch RzP

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Modulare Automation
und Prüftechnik

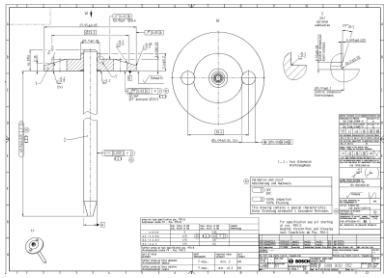
BOSCH
Technik fürs Leben

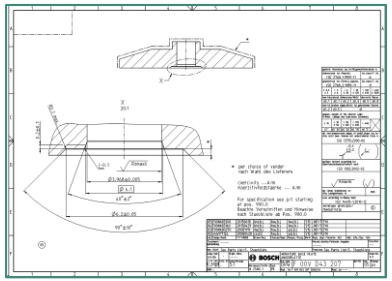
Gluth SYSTEM TECHNIK

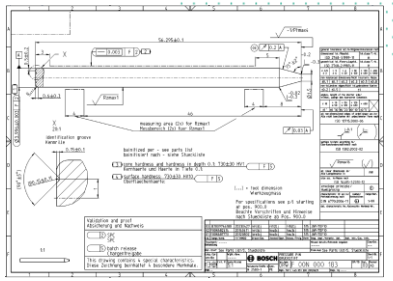
Products solenoid armature

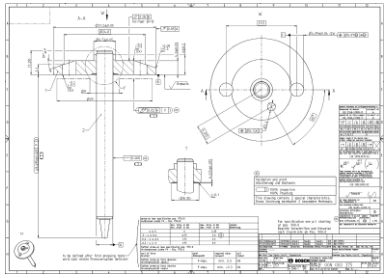
Parts name	Assembled part reference	Plate reference	Pin reference
Solenoid armature	F 00N 000 182	F 00V D43 207 Ø27,95mm, h5,4mm	F 00N 000 183 Ø4mm, L56,295mm
	F 00N 000 375	F 00N 000 274 Ø31,2mm, h6mm	F 00N 000 379 Ø5mm, L56,785mm
	F 00V D43 203	F 00V D43 200 Ø27,95mm, h5,4mm	F 00V D44 200 Ø4mm, L55,8mm
	F 00V D43 208	F 00V D43 207 Ø27,95mm, h5,4mm	F 00V D44 200 Ø4mm, L55,8mm

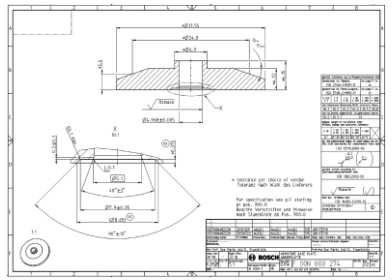


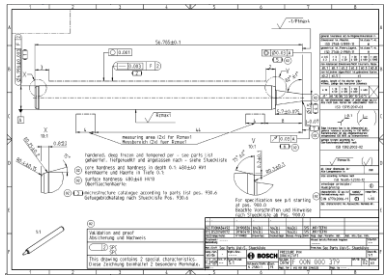


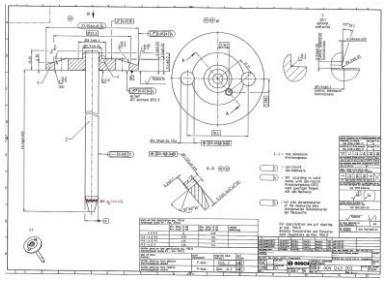


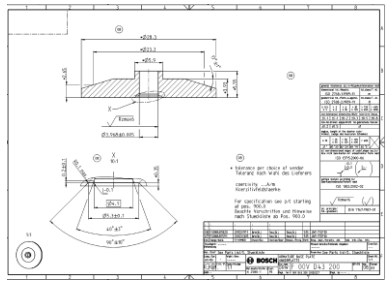


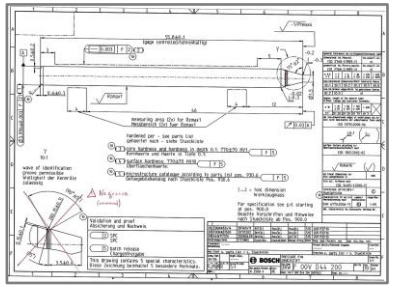


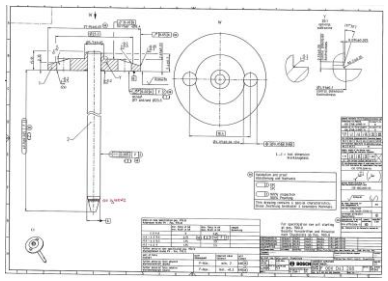


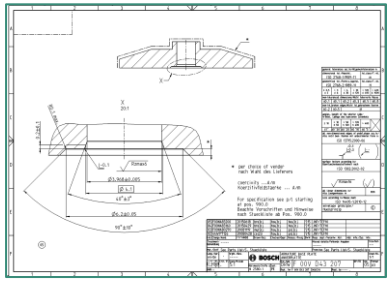


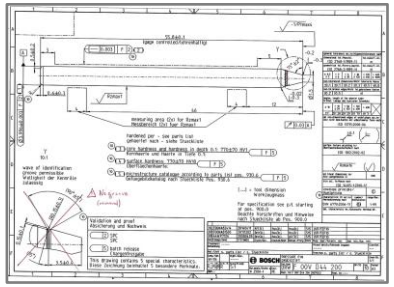












Cycle time: 8,2s

Basis of our offer:

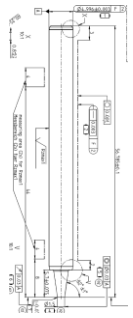
- Your specification ‚CDC solenoid armature assembly V1‘ from 02.04.2024
- Your specification ‚CDC solenoid armature assembly V2.1‘ from 25.06.2024
- Your process parameters F 00V D39 030 & F 00V D39 031
- Phone call Brault/Bauguil on 07.06.2024
- Former concept of M.Waas

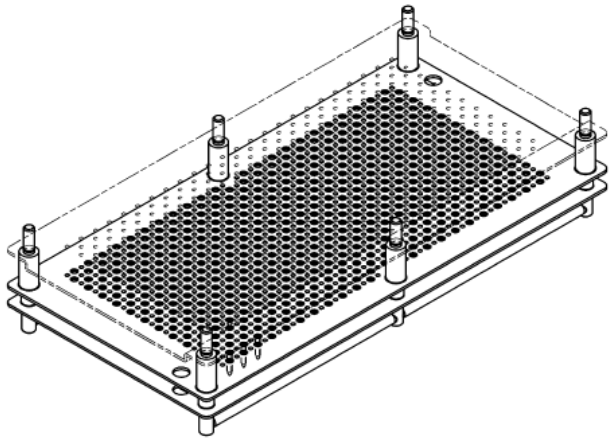
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loading & unloading trays

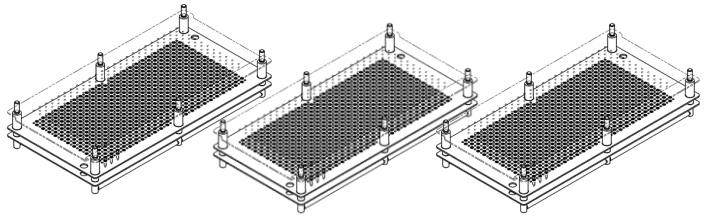
Tray pin

ET_0746W004





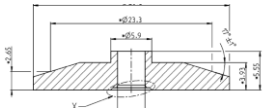
two trays per class, three classes
alarm & light, if tray empty
changing empty tray doesn‘t stop the machine

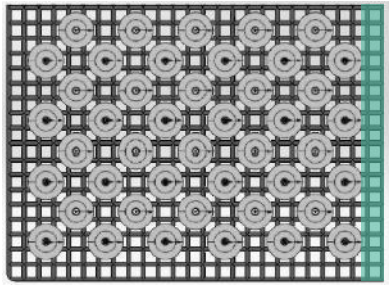


Tray plate

To be defined

- Tray aus plastic
- 44 parts, 329x238 mmm
- Parts on the B-surface in tray
- Ø3,9mm centering pin for all kind of plates
- Poka Yoke on the right side
- Compatible with conveyor





Tray solenoid armature

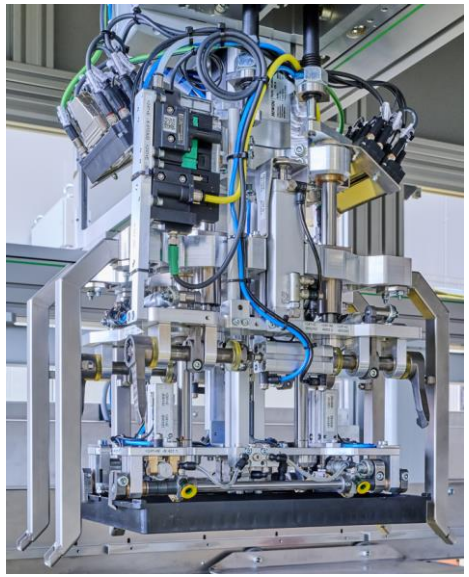
KLT 4280, Trays 6000.155.893

KLT 4280



Trays 6000.155.893

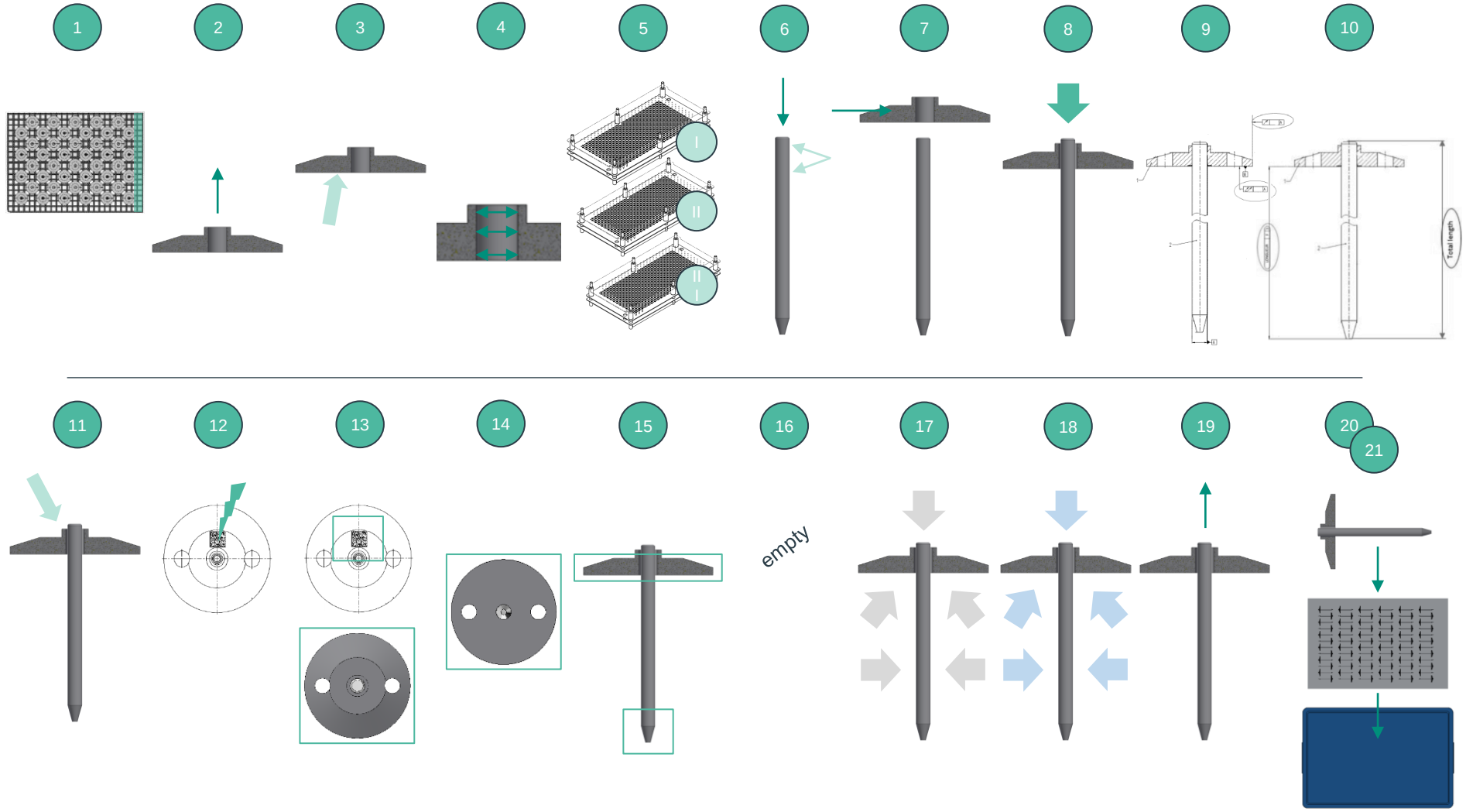
- We don‘t have any information about your Tray
- We assume that the tray can be automated and can be traded automatically in the KLT
- We have experience with similar technology for Bosch Rexroth Elchingen
- Picture shows the multifunctional gripper for Bosch Rexroth Elchingen
  - Grip KLT
  - Grip and handle “black palett” inside KLT
  - Grip and place cover on KLT
  - The gripper has a pneumatic spring system for
    - Exact positioning on pallet-stack
    - Emergency stop situation for potential collision



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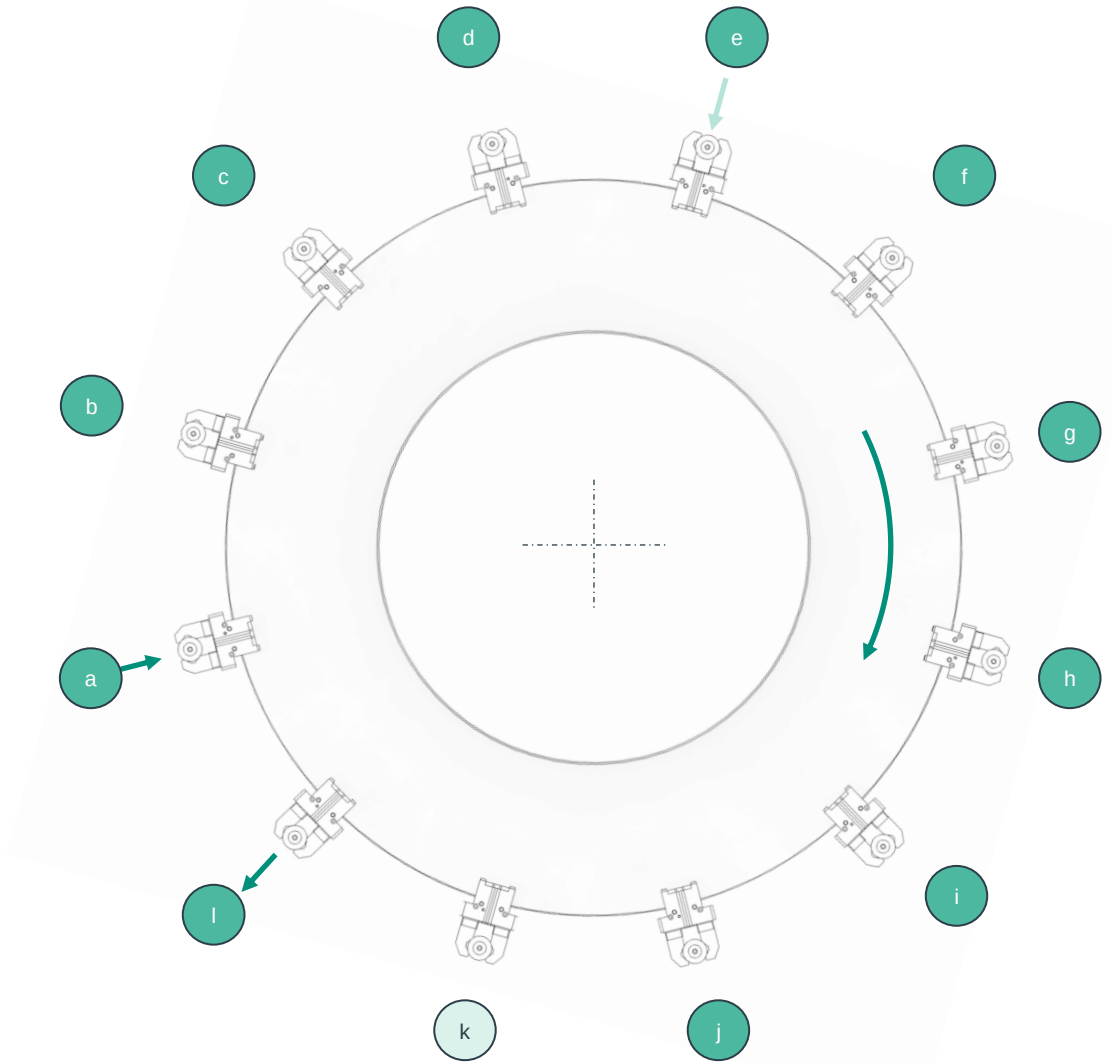
Required Processes V2.1

	Processes
1	feeding trays plate
2	loading plate
3	plate brushing
4	hole plate measurement
5	feeding trays pins
6	selection needle, oiling & loading in assembly tool
7	loading plate in assembly tool
8	press fitting
9	runout checking
10	length checking
11	back brushing
12	DMC marking (option)
13	DMC controlling (option) & AOI back
14	AOI front
15	AOI perimeter & pin tip
16	free place
17	cleaning solenoid armature (oil)
18	cleaning solenoid armature (compressed air)
19	unloading solenoid armature (iO & NiO)
20	output trays solenoid armature
21	output trays solenoid armature in KLTs



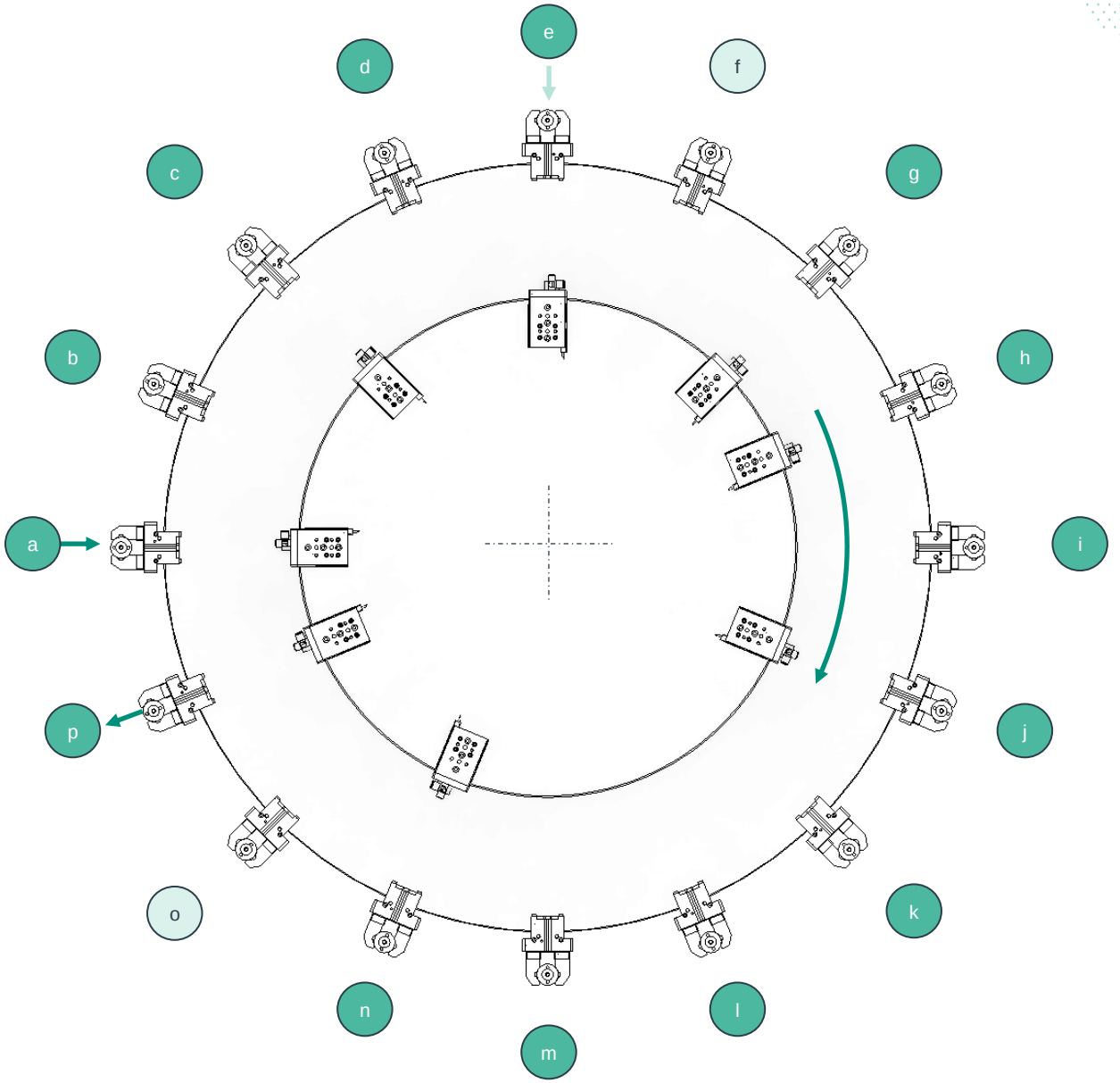
Transport unit (without options)

	Positions round table
a	loading plate
b	plate brushing
c	hole plate measurement
d	oiling inner hole plate / measered value to handling pins
e	loading pin in assembly tool & press fitting
f	runout checking
g	length checking
h	back brushing
i	AOI back & front
j	AOI perimeter & pin tip
k	free place
l	unloading solenoid armature (iO & NiO)



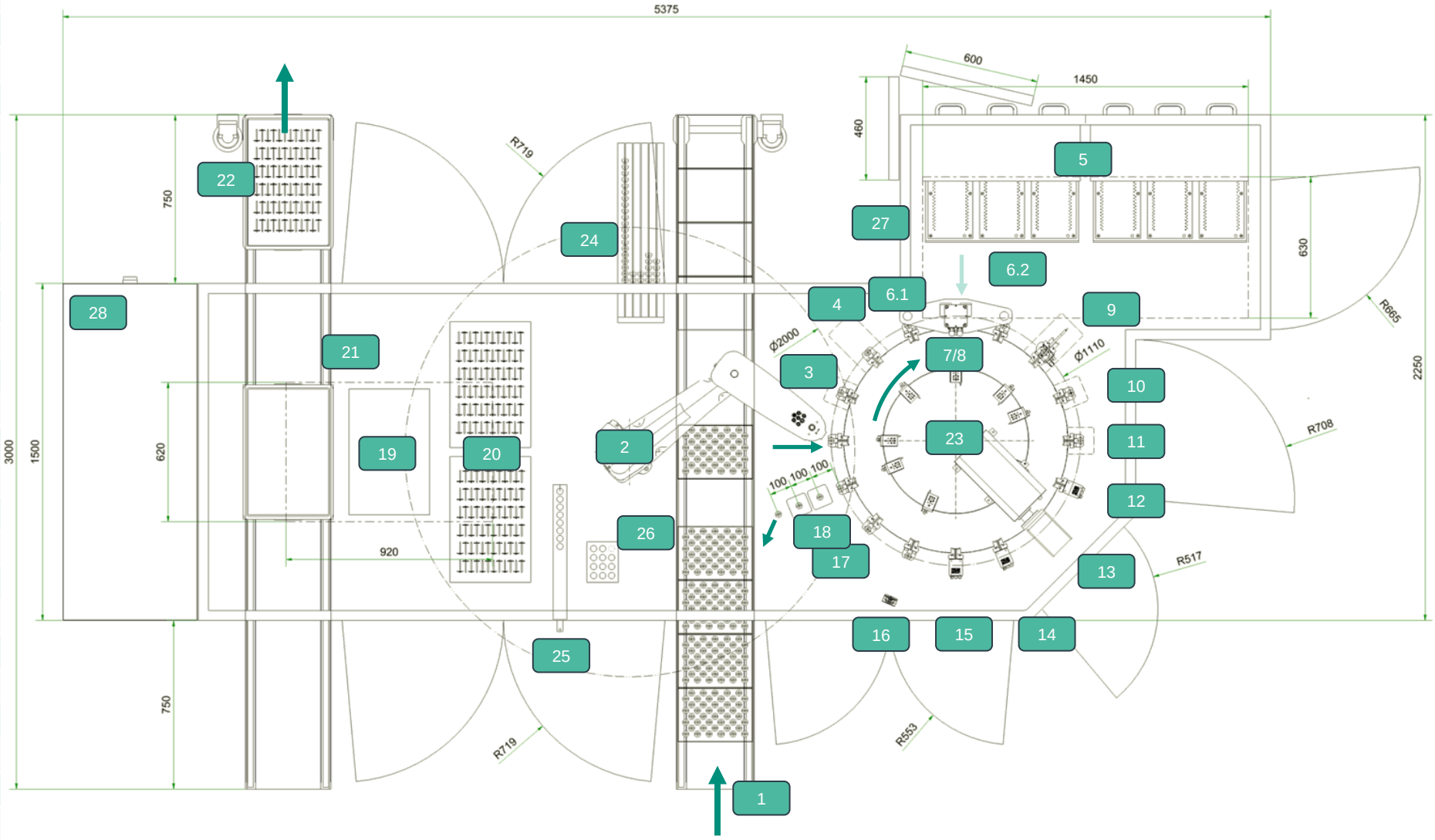
Transport unit (with options)

	Positions round table
a	loading plate
b	plate brushing
c	hole plate measurement
d	oiling inner hole plate / measered value to handling pins
e	loading pin in assembly tool & press fitting
f	free place
g	runout checking
h	length checking
i	back brushing
j	angle orientation solenoid armature (option)
k	DMC marking (option)
l	DMC controlling (option)
m	AOI back & front
n	AOI perimeter & pin tip
o	free place
p	unloading solenoid armature



Layout

	Stations
1	feeding trays plate
2	loading plate / unloading solenoid armature with Robot
3	plate brushing
4	hole plate measurement
5	feeding trays pins with 6 drawers
6.1	oiling inner hole plate / measured value to handling pins
6.2	selection needle & loading in assembly tool
7/8	loading plate in assembly tool & press fitting
9	runout checking
10	length checking
11	back brushing
12	angle orientation part for DMC marking (option)
13	DMC marking (option)
14	DMC controlling (option)
15	Possible position for „AOI back & front“
16	Possible position for „AOI perimeter & pin tip“
17	free place
18	handling solenoid armature with oiling/cleaning station
19	buffer pile empty trays solenoid armature
20	loading solenoid armatures in tray, 2x
21	handling trays solenoid armature
22	output KLTs with trays solenoid armature
23	transport unit / Round table
24	output NiO-Parts
25	drawer SPC-Parts
26	stabi-parts
27	base frame
28	controls & electrics
29	pneumatics



Differences to first offer A0759100A due to new RzP-requirements

Pos	Bezeichnung	Bemerkung
1	feeding trays plate	Station 1: Unchanged compared to Version A07591000A
2	loading Platte / unloading solenoid armature with robot	Station 2: Unchanged compared to Version A07591000A
3	plate brushing	Station 3: Unchanged compared to Version A07591000A
4	hole plate measurement	Station 4: Unchanged compared to Version A07591000A
5	feeding trays pins with 6 drawers	Station 5,6: 6 Tray-drawers instead of 3 trays inside machine
6	oiling inner hole plate / measured value to handling pins	Station 6.1: Complete new Station / requirement
7	selection needle & loading in assembly tool	Station 6.2: Unchanged compared to Version A07591000A
8	loading plate in assembly tool & press fitting	Station 7/8: Unchanged compared to Version A07591000A
9	runout checking	Station 9: Unchanged compared to Version A07591000A
10	length checking	Station 10: Additional measurement "total length"
11	back brushing	Station 11: Unchanged compared to Version A07591000A
12		
13		12,13,14: Option DMC-Marking
14		
15	AOI back & front	Station 15: Complete new Station / requirement However, only considered as a free station, as camera technology is completely unknown
16	AOI perimeter & pin tip	Station 16: Complete new Station / requirement However, only considered as a free station, as camera technology is completely unknown
17		
18	handling solenoid armature with oiling/cleaning station	Station 18: Complete new Station / requirement
19	buffer pile empty trays solenoid armature	Station 19: Complete new Station / requirement
20	loading solenoid armatures in tray, 2x	Station 20: Complete new Station / requirement
21	handling trays solenoid armature	Station 20: Complete new Station / requirement
22	output KLTs with trays solenoid armature	Station 22: Unchanged compared to Version A07591000A
23	transport unit / round table	Station 23: 12-part instead of 10-part round table
24	output NiO-Parts	Station 24: 5-part output-rails instead of 4 boxes
25	drawer SPC-Parts	Station 25: Unchanged compared to Version A07591000A
26	stabi-parts	Station 26: 11 instead of 3 Stabi-part-fixtures
27	base frame & protective structure	Bigger frame and protective structure
28	controls & electric	Additional expense due to additional stations
29	pneumatics	Additional expense due to additional stations

DMC-Marking: 16-part round table required

Comments / Notes

Station 13: Marking laser
- Warranty on the laser vmc3-class4 with special conditions between Bosch & Trumpf will stop at the end of 2025.
- No more Trumpf support or spare parts available after this date!

Station 4: Hole plate measurement
- Complete measurement equipment Plate-Inner-Ø (Stotz, measurement devices, pneumatics,…) incl. BCI-OpConPlus-Object is Bosch provision

Station 6: Oiling inner hole plate
- We recommend the oiling of the inner hole plate instead of oiling the pin

Station 15: AOI back & front
Station 16: AOI perimeter & pin
- Based on your shown „error patterns“ (your document „Visual inspection defect.pdf“) it is not possible to define the requiered camera-equipment
- Bosch feasibility study is requiered for definition the complete camera/lighting technology
- Without this information we cannot design / calculate this both stations
- That‘s why we only have on 2 empty stations available
- The complete 2 Stations (Camera- and Station- Hardware, complete engineering and so on) are not calculated

Station 16: AOI perimeter & pin
- Identical to Station 15

Station 18: Handling solenoid armature with oiling / cleaning station
- A „Pilgrim step handling“ position the parts
  - from Round table to Pos-1 „Oiling / cleaning station“
  - from Pos-1 to Pos-2 „Blow off station“
  - from Pos-2 to Pos-3 „Pick up part by Robot“
- We cannot guarantee that the cleanliness of the parts (required in the specifications) is guaranteed by the simple oiling process

OpConPlus Objects
- We assume that all required OpConPlus objects exist. If this is not the case, the commissioning of a new OpConPlus object is not within the scope of our offer

Price indication

Total price:

Complete machine: 1.490.000€

- *Without option*
- *Including re-commissioning in Rodez*

Option DMC-Marking: 108.000€

- *Including*
  - *16-fold round table instead of 12 fold-round table, incl. 4 additional clamping units and 1 additional opening unit (additional expenditure compared to first offer-version)*
  - *Angle Orientation before DMC-Marking*
  - *DMC-Marking*
  - *Control DMC*
- *Bosch provisions are:*
  - *Complete Trumpf laser for DMC-marking incl. OpConPlus-Object*
  - *Complete Cognex camera equipment for Angle Orientation*
  - *Complete Cognex camera equipment for Control*



Danke
für die Aufmerksamkeit

Gluth SYSTEM TECHNIK

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